Instituto Tecnológico Autónomo de México

Does Qualifying Really Matter? The Causal Impact of Making It to Q2 and Q3 on Formula 1 Race Outcomes

Research Paper

Investigación Aplicada I

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Introduction

How much does a tenth of a second matter? In the world of Formula 1, it can mean everything—especially when 20 drivers are on track, all vying to set the fastest time. This is the nature of Formula 1 qualifying, where just half a second or even less can be the difference between starting at the front of the grid or the back.

There is a common belief in Formula 1 that a poor qualifying performance on Saturday can ruin a driver's Sunday, making their race far more challenging as they struggle to climb through the field and recover lost positions. But is this belief really true?

The core question this paper addresses is whether the importance of qualifying in Formula 1 is truly as significant as believed: Does making it to Q2 and Q3 impact a driver's race performance? This study will examine how advancing to Q2 and Q3 affects various race outcomes, including grid position, final race position, the likelihood of a DNF (Did Not Finish), and tyre strategy, in order to assess the overall impact on race performance.

This paper employs a Regression Discontinuity Design (RDD) as the empirical strategy to identify the causal effect of qualifying on race outcomes. The rationale for using RDD in this context is that drivers who either just barely advanced to the next qualifying session or narrowly missed out are likely to be comparable. By focusing on these

marginal cases, we can isolate the impact of qualifying on subsequent race performance.

The analysis identifies significant effects of qualifying performance on race outcomes, with advancing to Q2 producing the largest impacts. Progressing to Q2 and Q3 positively affects both Grid Position and Final Race Position. However, no significant effects are observed on the use of Alternative Strategies or the number of pit stops.

This paper aims to contribute to the sports economics literature, particularly in the context of Formula 1, where the analysis of causal effects remains underexplored. By applying a method such as Regression Discontinuity Design, which has been widely used in other settings, including sports, this study seeks to address the gap in understanding causal relationships within Formula 1.

Chapter 1 talks about the relevant literature and related work to this paper. Chapter 2 gives an overview of Formula 1, its qualifying format, and the teams' performance across a race weekend. Chapter 3 delves into the world of Regression Discontinuity Design as the empirical strategy for this analysis and explains why it is useful in the context of Formula 1 qualifying. Like in this analysis, Formula 1 would not function without data; therefore, Chapter 4 highlights and describes the data used for this study. Finally, in Chapter 5, the main results of the analysis are discussed.

Chapter 1

Related Work

1.1 Formula 1 Qualifying Impact

As previously mentioned, there is a widely held belief that qualifying plays a crucial role in a driver's overall race weekend. This belief has driven research into the impact of qualifying on race performance.

Heilmeier et al. (2020) implement neural networks to calculate optimal race strategies. Their model incorporates the starting position as a key parameter to determine race strategy. Beyond identifying the optimal strategy, they find that starting further back on the grid increases race time, which is precisely the factor their approach seeks to minimize when determining the best strategy. Similarly, McCarthy and Rotthoff (2013) employ a Probit model to assess the effect of starting position on the probability of first-lap collisions. Their results show that starting in 10th position is the most dangerous for first-lap contact, while the safest positions are pole or near the back of the grid to avoid collisions. Finally, Patil et al. (2022) run a linear regression (OLS) on various race outcomes and find that, on average, qualifying one spot lower (e.g., from 9th to 10th) results in a loss of 3.44 points

in the season's total.

All of these previous works demonstrate that qualifying has a significant effect on various race outcomes; however, these effects are correlational, and the causal effects of qualifying remain underexplored.

1.2 Regression Discontinuity Design in Other Sports

While the causal effects of qualifying in Formula 1 are underexplored, similar analyses have been conducted in other sports, particularly using Regression Discontinuity Designs (RDD).

Speer (2023) implements a Fuzzy RDD to examine the effects of relegation or promotion on team performance outcomes in football teams across various European leagues. By comparing teams that were just relegated with those that narrowly avoided relegation, and similarly, teams that were just promoted to the top league with those that just missed promotion, the study finds long-term effects (3 to 4 years) on team performance. Teams that were just relegated struggle to improve their performance enough to return to the top league, while teams that were just promoted are unlikely to be relegated again within the next 3 to 4 years.

Klein Teeselink et al. (2023) analyze the effect of a team being marginally behind at halftime on the probability of winning the game. They examine four sports: American Football, Australian Football, Rugby, and Basketball, using the halftime score as the running variable. Their findings show no significant effect of being marginally behind at halftime on the likelihood of winning the match.

Regression Discontinuity Design (RDD) has been successfully

applied in various other sports, demonstrating its effectiveness in isolating causal effects in competitive settings. The use of RDD in sports like football and basketball provides strong evidence that it can be a powerful tool for analyzing the impact of qualifying on race outcomes in Formula 1.

1.3 Outside Formula 1

In Formula 1, external factors beyond the control of drivers or teams can influence qualifying outcomes. Similarly, outside of Formula 1, exogenous variations can shape individuals' life trajectories. For example, receiving a scholarship or financial assistance for education can create opportunities that significantly affect long-term outcomes.

Landaud et al. (2022) analyze the impact of exam luck on educational attainment and labor market success in Norway. They identify random variations in the content of high school exams as a source of exogenous variation. Their findings suggest that "lucky" students who take exams on subjects they excel in achieve higher GPAs and are more likely to earn a diploma. To measure this effect, they construct a variable called "GPA luck," which they use as an instrumental variable to estimate the causal impact of exam outcomes on long-term success. The results indicate that exam-induced variations in GPA significantly affect the likelihood of achieving a good GPA, with "GPA luck" also translating into higher wages in the labor market.

Similarly, Zimmerman (2014) employs a RDD to study the effect of college admission on earnings. The study finds that students just above the cutoff for the high school grades required for college admission earn, on average, $372 more per quarter than those just below the cutoff.

These findings demonstrate that when advantages arise from characteristics such as grades—which can be influenced by exogenous factors—"luck" plays a significant role in determining outcomes. This luck can provide individuals with advantages or disadvantages, ultimately influencing their opportunities and achievements in ways beyond their control.

Chapter 2

Context

2.1 Formula 1 (F1) Overview

Formula 1 (F1) is the pinnacle of motorsport, featuring 20 drivers across 10 teams. Each team fields two cars, contributing to both the Constructors' Championship and the Drivers' Championship. Occasionally, there are more than 20 drivers in the standings due to reserve drivers, who step in when a regular driver is injured, ill, or otherwise unable to participate in a race. These reserve drivers accumulate points for themselves in the Drivers' Championship but also contribute to their team's tally in the Constructors' Championship. For example, during the Saudi Arabian Grand Prix, Carlos Sainz suffered from appendicitis, and Oliver Bearman was called in to substitute for him. Bearman also made another appearance during the Azerbaijan Grand Prix, filling in for Kevin Magnussen, who was serving a race ban. Table 2.1 and Table 2.2 show the current standings in both the Drivers' and Constructors' Championships, respectively.

The first Formula 1 season began in 1950 at Silverstone, which

Position	Driver	Points	Team
1	Max Verstappen	354	Red Bull Racing
2	Lando Norris	297	McLaren
3	Charles Leclerc	275	Ferrari
4	Oscar Piastri	247	McLaren
5	Carlos Sainz	215	Ferrari
6	Lewis Hamilton	177	Mercedes
7	George Russell	167	Mercedes
8	Sergio Pérez	150	Red Bull Racing
9	Fernando Alonso	62	Aston Martin
10	Nico Hülkenberg	29	Haas
11	Lance Stroll	24	Aston Martin
12	Yuki Tsunoda	22	RB
13	Alexander Albon	12	Williams
14	Daniel Ricciardo	12	RB
15	Kevin Magnussen	8	Haas
16	Pierre Gasly	8	Alpine
17	Oliver Bearman	7	Ferrari/Haas
18	Franco Colapinto	5	Williams
19	Esteban Ocon	5	Alpine
20	Liam Lawson	2	RB
21	Zhou Guanyu	0	Kick Sauber
22	Logan Sargeant	0	Williams
23	Valtteri Bottas	0	Kick Sauber

Table 2.1. 2024 Drivers' Championship until US GP

remains a part of the calendar today. An F1 season typically runs from February or March to November, occasionally extending into

Position	Team	Points	Wins	Podiums
1	McLaren	544	5	18
2	Red Bull	504	7	16
3	Ferrari	496	4	16
4	Mercedes	344	3	7
5	Aston Martin	86	0	0
6	Haas	38	0	0
7	RB	36	0	0
8	Williams	17	0	0
9	Alpine	13	0	0
10	Kick Sauber	0	0	0

Table 2.2. 2024 Constructors' Championship until US GP

December, and includes between 20 to 24 races. Usually, there is a summer break after 2 or 3 races following the midpoint of the season, during which teams shut down their factories and take a holiday. This break is essential because teams spend the remaining time between races and the winter break between seasons focused on improving their cars. Table 2.3 shows the 2024 F1 calendar.

The Formula 1 weekend begins with three one-hour Free Practice sessions, where teams test car setups and gather data. Free Practice 1 and 2 take place on Friday, while Free Practice 3 occurs on Saturday before Qualifying. Just before Qualifying, cars enter *Parc fermé* conditions, during which the car's core setup is fixed. During this time, teams can only perform limited maintenance, such as replacing a front wing, but no modifications to the car's setup or suspension are allowed. Violations result in the car starting from the pit lane.

Qualifying is held on Saturday to determine the starting grid. It

Round	Grand Prix	Circuit	Date
1	Bahrain GP	Bahrain International Circuit	2 Mar
2	Saudi Arabian GP	Jeddah Street Circuit	9 Mar
3	Australian GP	Albert Park	24 Mar
4	Japanese GP	Suzuka Circuit	7 Apr
5	Chinese GP	Shanghai International Circuit	21 Apr
6	Miami GP	Miami International Autodrome	5 May
7	Emilia Romagna GP	Autodromo Enzo e Dino Ferrari	19 May
8	Monaco GP	Circuit de Monaco	26 May
9	Canadian GP	Circuit Gilles Villeneuve	9 Jun
10	Spanish GP	Circuit de Catalunya	23 Jun
11	Austrian GP	Red Bull Ring	30 Jun
12	British GP	Silverstone Circuit	7 Jul
13	Hungarian GP	Hungaroring	21 Jul
14	Belgian GP	Spa-Francorchamps	28 Jul
15	Dutch GP	Circuit Zandvoort	25 Aug
16	Italian GP	Autodromo Nazionale Monza	1 Sep
17	Azerbaijan GP	Baku City Circuit	15 Sep
18	Singapore GP	Marina Bay Street Circuit	22 Sep
19	United States GP	Circuit of the Americas	20 Oct
20	Mexican GP	Autodromo Hermanos Rodriguez	27 Oct
21	Brazilian GP	Autodromo Jose Carlos Pace	3 Nov
22	Las Vegas GP	Las Vegas Street Circuit	24 Nov
23	Lusail GP	Losail International Circuit	1 Dec
24	Abu Dhabi GP	Yas Marina Circuit	8 Dec

Table 2.3. F1 2024 Calendar

consists of three knockout sessions—Q1, Q2, and Q3—with the fastest driver earning pole position for the race. Details on qualifying are elaborated later.

The main event, the race, takes place on Sunday and lasts between

1.5 to 2 hours, depending on the circuit and race conditions. Cars are required to make at least one pit stop to change tyres. Points are awarded to the top ten finishers.

Throughout the Formula 1 season, there are special Grand Prix weekends known as Sprint Weekends. These weekends were introduced to reduce the number of practice sessions and to enhance the excitement and unpredictability of the racing experience.

In a Sprint Weekend, the schedule is condensed, featuring only one Free Practice session instead of the usual three. This is followed by a Sprint Qualifying session, which functions similarly to the qualifying for the main race and determines the starting grid for the Sprint Race. The Sprint Race, held on Saturday morning, is a 100 km event run from lights to flag and does not require a mandatory pit stop. Consequently, teams must strategically manage their tyre performance throughout the race, as making a set of tyres last until the end is crucial for achieving a favorable result. Three hours after the Sprint concludes, Qualifying for the main race takes place, with the main race occurring as usual on Sunday.

Despite the unique format of Sprint Weekends, this paper primarily focuses on the traditional qualifying sessions and the main races.

2.2 How does F1 qualifying work?

Pirelli, the sole tyre supplier in Formula 1, has developed five compounds of dry tyres, designated as C1, C2, C3, C4, and C5, ranging from the hardest to the softest. Typically, softer compounds provide faster lap times but offer less durability. For wet conditions, there are two types of tyres: Intermediates and Full Wets. Pirelli decides which compounds will serve as the Soft, Medium, and Hard tyres for each race weekend,

based on the unique characteristics of the circuit.

At the beginning of the weekend, each driver receives a total of 13 new sets of tyres, known as the tyre allocation. This allocation consists of 2 sets of Hard tyres, 3 sets of Medium tyres, and 8 sets of Soft tyres. Additionally, drivers are provided with 5 sets of Intermediate tyres and 2 sets of Full Wet tyres. Teams must strategically manage their tyre allocation across Free Practices, Qualifying, and the race. New tyres generally offer better performance than used ones, necessitating strategic decisions from the first hour of practice on Friday. The majority of the Soft tyre sets are typically reserved for Qualifying, as they are the fastest compound, although teams may opt to run some fast laps on Softs during practice for qualifying simulations.

At the start of the weekend, the track is considered *green*, meaning the racing line—the optimal path for cars to achieve the best lap times—has minimal grip. As cars repeatedly traverse this line, they lay down rubber from their tyres, enhancing grip on the track. Consequently, lap times tend to improve as the session progresses due to this increased grip, a phenomenon known as track evolution. While track evolution on permanent racing circuits may be moderate between Free Practice 1 and the end of the race, street circuits can experience dramatic changes even within a single session.

With track evolution, the conditions of the circuit improve as the session progresses, meaning that the optimal strategy for teams is to set their best lap times at the end of the session when the track is at its peak performance. Notably, even when the clock reaches zero, if a driver crosses the starting line before time expires, they are still permitted to complete their lap. However, this strategy carries risks, as teams may encounter traffic, crashes, driver mistakes, or yellow and red flags, which can prevent them from finishing their laps. This is particularly crucial

for drivers attempting to escape the knockout zone during Qualifying; if they experience any of these issues during their final attempt, they may not have enough time to make another lap, potentially leading to their elimination from the qualifying session.

For example, the Azerbaijan Grand Prix takes place at the Baku City Circuit, a street circuit that winds through the streets of Baku. During this year's qualifying session in Q1, there was significant track evolution, prompting everyone to try to set their fastest laps at the end of the session. McLaren found themselves with both drivers in the elimination zone at this critical moment, despite having one of the fastest cars on the grid. Just as Lando Norris was finishing his lap, a yellow flag was deployed due to an incident in the same sector, forcing him to lift off the throttle and preventing him from completing his lap. As a result, he was eliminated in Q1, while his teammate, Oscar Piastri, who was able to finish his lap, qualified second and went on to win the race on Sunday.

Track evolution is not the only factor to consider during Qualifying; tyre allocation presents another challenge for teams. New soft tyres can significantly impact lap times, potentially making the difference between achieving pole position and qualifying in fourth place. However, teams have a limited number of new soft tyres, forcing them to carefully decide how many laps to run in each qualifying session.

While drivers theoretically can complete as many laps as they want while time remains in the session, they are practically constrained by their tyre allocation. Teams with faster cars may opt to use only one set of new tyres in Q1 and Q2, preserving two sets of new softs for the pole shootout in Q3. Consequently, their banker lap—the first lap they complete to ensure they have a time on the board—must be good enough to advance them to the next qualifying session.

The issue here is similar to the risks associated with track evolution: traffic and driver mistakes during their banker lap could result in top teams being eliminated early in qualifying. This can happen if teams are overly confident and choose to save a set of tyres for later use. The dilemma is straightforward: teams can either use a fresh set of tyres to secure a solid lap and ensure progression to the next session, which may put them at a disadvantage during the pole shootout, or they can take the risk and save the tyres for the best chance at pole position. This is why drivers usually approach qualifying as if each lap is their only opportunity to set a competitive time.

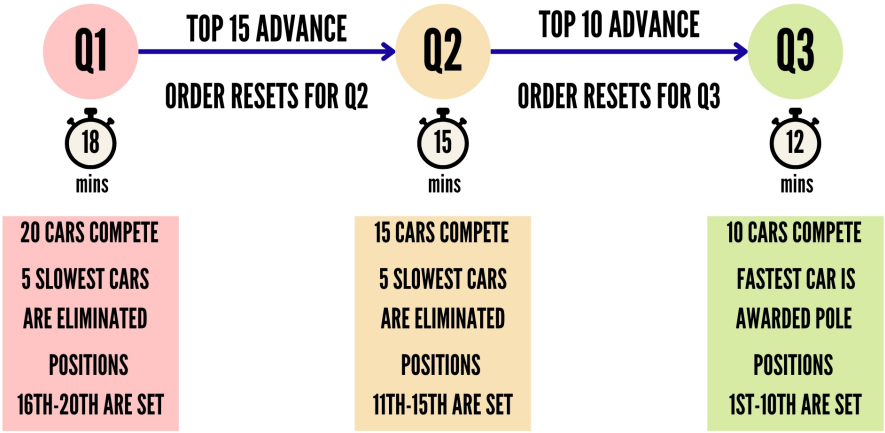

Figure 2.1. Formula 1 Qualifying Format

Figure 2.1 describes the current qualifying format for each race weekend. Qualifying usually lasts for an hour, each session with its allocated time with breaks between session lasting usually 5 minutes. If there is an incident on the track that needs to be cleared before the other session starts like an accident or debris the breaks between sessions can last longer. Each session is independent from each other, so the results or the order from the previous session do not carry over.

In Q1, all 20 drivers participate, each trying to set the fastest time possible before the 18 minutes are up. This is the most frantic session of all the qualifying, with 20 cars on track. Teams often find it difficult to slot their car into an optimal gap to avoid traffic on their fast lap. The objective here is to set a faster time than the 15th fastest time, as the 16th to 20th fastest times are eliminated at the end of the session, and these positions are set.

In Q2, the top 15 drivers from Q1 compete for a place in the final qualifying session. This session lasts 15 minutes and shares the same objective: to set a time faster than the knockout threshold to avoid elimination. The top 10 drivers from Q2 advance to Q3, while positions 11th to 15th are finalized based on their best lap times.

Q3 is the shootout for pole position, where the top ten drivers compete for the coveted spot. In this session, drivers do not need to worry about saving a set of new tyres for subsequent sessions; it is an all-out effort where drivers typically aim to complete two fast laps, each on a new set of soft tyres. However, they still want to ensure that their banker lap is competitive enough to secure a good starting position. At the end of the session, the driver with the fastest laptime is awarded pole position, and the final grid positions from 1st to 10th are determined.

For example, during this year's US Grand Prix, Lando Norris

found himself on provisional pole after the first round of laps. However, as all drivers started their second laps, Max Verstappen and Carlos Sainz were quicker than Norris in the first sector. Just as they were setting their best times, George Russell crashed, bringing out the yellow flags. Consequently, no one could complete their laps, allowing Norris to secure pole position.

2.3 Team Performance in Qualifying

This section provides an overview of the performance of current Formula 1 teams during qualifying sessions. The variables *MadeItToQ2* and *MadeItToQ3* serve as key treatment variables in this analysis, highlighting the importance of understanding how teams have fared in qualifying over recent years.

Team	**Q2 (%)**	**Q3 (%)**	**Top 3 (%)**	**Pole (%)**
Mercedes	98.28	95.18	49.56	15.79
Red Bull Racing	96.49	91.82	44.25	18.58
Ferrari	98.27	86.55	36.40	14.04
McLaren	91.77	80.48	11.84	1.32
Alpine	86.52	57.79	1.78	0
Aston Martin	71.62	54.94	4.41	0.44
RB	76.65	46.15	0.44	0
Haas F1 Team	50.22	42.34	0	0
Kick Sauber	54.39	27.87	0	0
Williams	34.82	22.97	0.88	0

Table 2.4. Team qualifying performance (2019–2024) (Ordered by Q3 (%))

Table 2.4 presents the percentages of times each team has successfully progressed to Q2 and Q3, as well as the Top 3 and Pole percentages, across all qualifying sessions from 2019 to 2024. These metrics illustrate how frequently each team has advanced to these critical stages with at least one of their drivers.

Mercedes and **Ferrari** demonstrate the highest Q2 percentages, indicating consistent performance in qualifying sessions. Specifically, Mercedes leads with a Q2 percentage of 98.28%, closely followed by Ferrari at 98.27%. This suggests strong overall qualifying performance and reliability from both teams.

In terms of Q3 performance, **Mercedes** and **Red Bull Racing** emerge as the top performers. Mercedes boasts an impressive Q3 percentage of 95.18%, while Red Bull Racing closely follows with 91.82%. These figures indicate that both teams not only frequently reach Q3 but also exhibit a high success rate in advancing through the qualifying stages to secure optimal starting positions.

Interestingly, while **Red Bull Racing** has the highest pole percentage (18.58%), it does not have the same success rate in Q3 as **Mercedes**. This suggests that Red Bull may be adopting a more aggressive strategy, opting to conserve new sets of tyres for the pole position fight rather than ensuring a guaranteed place in Q3.

Conversely, teams like **Williams** and **Haas F1 Team** show the lowest percentages for Q2, with **Kick Sauber** struggling in Q3. Williams achieves a Q2 percentage of only 34.82% and a Q3 percentage of 22.97%, while Haas records a Q2 percentage of 50.22% and a Q3 percentage of 42.34%. This performance highlights significant challenges faced by these teams in consistently qualifying for the final stages of the session.

2.4 Team Performance in Race

To analyze the effect of advancing through Q1 and Q2 on race outcomes, it is essential to examine the performance of teams in races as well.

Table 2.5 displays the average grid position, average race finishing position, and the percentage of Did Not Finish (DNF) incidents for each team across all races from 2019 to 2024.

Team	**Mean Grid Position**	**Mean Race Position**	**DNF (%)**
Mercedes	4.78	5.24	7.76
Red Bull Racing	5.52	5.47	11.21
Ferrari	6.00	7.07	12.07
McLaren	8.55	9.10	10.78
Alpine	10.91	10.80	15.52
Aston Martin	11.89	10.90	13.79
RB	12.55	12.36	15.52
Kick Sauber	14.54	13.68	13.36
Haas F1 Team	14.62	15.04	18.97
Williams	16.00	15.28	20.35

Table 2.5. Team race performance (2019-2024) (Ordered by Mean Race Position)

Teams that achieve high percentages in Q2 and Q3, such as **Mercedes** and **Red Bull Racing**, demonstrate strong race performances, reflected in their superior average grid positions, finishing positions, and low DNF percentages.

Conversely, teams with the lowest Q2 and Q3 percentages, such as **Williams** and **Haas F1 Team**, tend to exhibit higher DNF percentages alongside poor average grid and race positions.

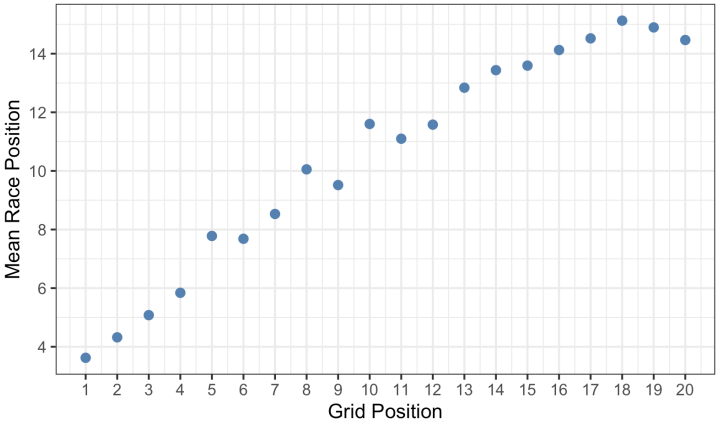

Figure 2.2. Mean Race Position by Grid Position

Figure 2.2 illustrates the mean final race position based on grid position. A positive correlation is observed, indicating that drivers who start further back tend to finish further back, while those starting closer to pole position (1st place) finish nearer the front.

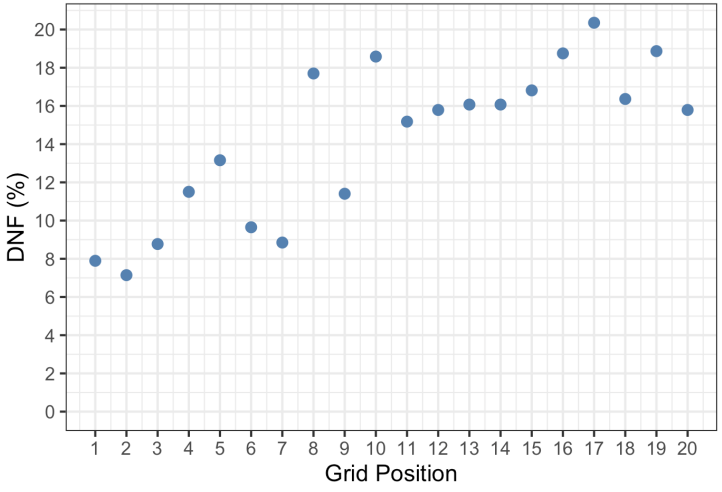

Figure 2.3. DNF Percentage by Grid Position

Figure 2.3 presents the DNF percentage associated with each grid position. Although the trend is less pronounced compared to the mean final race position, it can be noted that starting in the rear, particularly in 17th place, correlates with the highest DNF rates. Conversely, drivers starting at the front have lower DNF rates, although drivers in 6th and 7th positions also show lower DNF rates compared to those in 4th or 5th.

All these observations suggest a potential effect of the treatment (advancing through Q1 and Q2 into the final session of qualifying) on race outcomes; however, further analysis is necessary to substantiate this relationship.

2.5 Tyre Strategy and Number of Pitstops

An important aspect of this analysis is the tyre strategy used during races. Formula 1 employs five different types of tyres or compounds: three for dry conditions (Soft, Medium, and Hard) and two for wet conditions (Intermediates and Full Wets).

The Soft compound is the fastest but degrades quickly, losing performance after just a few laps. The Medium compound provides a balance between performance and durability, while the Hard compound, although generally the slowest, is the most durable of the three.

Table 2.6 summarizes the starting tyre percentages for each type of compound, the alternative strategy percentage for drivers who opted for a different strategy than the most popular one, and the average number of pitstops based on grid position.

The highest percentage of Soft compound usage occurs at 7th place, while the lowest is seen at 14th. Conversely, the highest percentage of

Grid Position	Soft (%)	Medium (%)	Hard (%)	Alternative Strategy (%)	Avg Number of Pitstops
1	32.456	56.140	0.877	62.281	1.096
2	27.679	60.714	2.679	59.821	1.098
3	34.211	54.386	0.877	59.649	1.088
4	37.168	49.558	3.540	62.832	1.097
5	36.842	52.632	0.000	60.526	1.053
6	37.719	50.000	2.632	68.421	1.053
7	45.133	41.593	4.425	70.796	1.168
8	41.593	41.593	6.195	69.027	1.088
9	47.368	37.719	4.386	67.544	1.070
10	42.478	44.248	3.540	69.912	1.080
11	26.786	49.107	14.286	73.214	1.170
12	21.930	56.140	11.404	75.439	0.982
13	24.107	51.786	14.286	75.893	1.134
14	21.429	61.607	8.929	76.786	1.080
15	27.434	54.867	7.080	73.451	1.142
16	26.786	52.679	12.500	81.250	1.152
17	28.319	47.788	13.274	80.531	1.053
18	22.727	56.364	11.818	86.364	1.109
19	21.698	47.170	19.811	86.792	1.019
20	15.789	49.624	23.308	87.970	1.068

Table 2.6. Tyre Strategy (2019-2024)

Medium compound usage is at 14th place, with the lowest percentages observed at 7th and 8th positions. For the Hard compound, the highest usage is at 20th place, while the lowest, at 5th place, shows 0%.

While the number of pitstops does not significantly vary with grid position, as teams generally aim to minimize time spent in the pits, the predominant tyre strategy varies based on starting position. This is especially true for drivers starting toward the back of the grid, who often adopt different strategies to gain positions.

For instance, if a driver experiences a poor qualifying session, the team may choose a different approach compared to others. This could involve starting on a Hard compound to create a longer first stint—allowing them to gain positions while other drivers pit—or opting for a Soft tyre, which provides better grip for overtaking at the race start.

Overall, this analysis indicates that tyre strategy is closely tied to grid position so making it to Q2 or Q3 may have a significant effect on this outcome.

Chapter 3

Empirical Strategy

3.1 Why Use Regression Discontinuity Design?

As explained by Imbens and Lemieux (2008), Regression Discontinuity Design (RDD) is highly effective when a treatment assignment is based entyrely or partially on whether a covariate crosses a particular threshold, referred to as the running or forcing variable. The key assumption in RDD is that the relationship between the running variable and the outcome of interest is smooth. Therefore, any abrupt change or discontinuity at the cutoff point is indicative of a causal effect of the treatment.

In the context of Formula 1 qualifying, RDD is particularly well-suited due to the knockout format. Drivers must set lap times below a certain threshold to advance to the next qualifying session. By using the gap to the knockout time as the running variable (standardized for comparability), an RDD analysis becomes feasible.

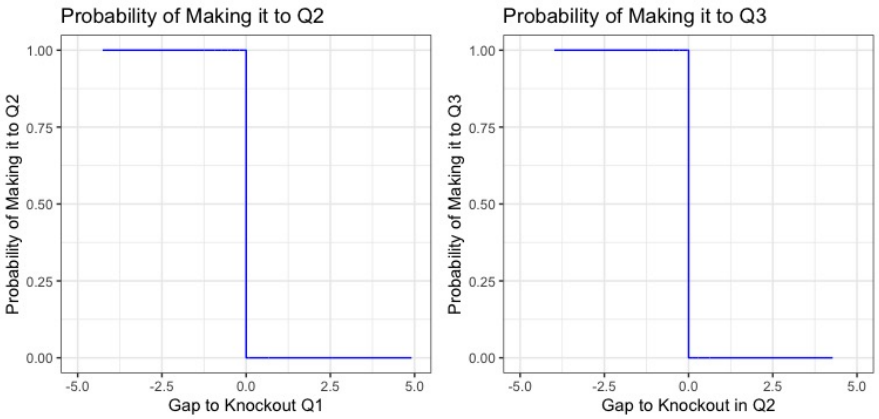

Figure 3.1. Assignment probabilities.

Note: The figure shows the treatment variables MadeItToQ2 and MadeItToQ3 plotted against their respective gaps to knockout in the previous session. At the cutoff, the probability of treatment jumps from 0 to 1.

As shown in Figure 3.1, this analysis qualifies as a Sharp RDD. The probability of progressing to Q2 or Q3 changes abruptly from 0 to 1 at the cutoff (knockout time), where drivers slower than the threshold are eliminated, and those faster advance.

3.2 Estimation

The estimation strategy for assessing the effect of advancing to Q2 and Q3 on the outcome variable is specified as follows:

$$Outcome_i = \tau_{SRD}\text{MadeItToQ2/Q3}_i + \alpha_0\text{GapToKnockout}_i + \alpha_1(\text{GapToKnockout}_i \times \text{MadeItToQ2/Q3}_i) + \lambda_{\text{circuit-team}_i} + \epsilon_i \quad (3.1)$$

In this model, τ_{SRD} represents the estimated causal effect of the

treatment, defined as making it to Q2 or Q3. The variable MadeItToQ2/Q3 serves as the treatment indicator, depending on whether the effect of advancing to Q2 or Q3 is being estimated.

The running variable, GapToKnockout, is centered around a cutoff value of 0, representing the knockout threshold where teams faster than this value advance to the next session, while teams slower are eliminated from qualifying.

Fixed effects, $\lambda_{\text{circuit-team}}$, are included for each circuit-team pairing. These fixed effects control for variations in team performance across circuits, as some circuits may suit certain team strengths, such as high-speed sections or complex technical segments. Although the calendar order may change slightly from year to year, these circuit-team effects also serve to control for temporal factors affecting performance.

3.3 Identifying Assumptions

The assumptions of a Regression Discontinuity Design (RDD) relate to continuity. First, potential outcomes must be continuous in *GapToKnockout* around the cutoff value of *GapToKnockout* $= 0$. Second, other covariates should also remain continuous at the cutoff. This implies that the characteristics of drivers near the cutoff on either side must be comparable, indicating there should be no discontinuity in the covariates at the cutoff. The only factor that should change at the cutoff is the assignment to treatment—specifically, qualifying for the next session, which means there are no compound effects involved.

The no-sorting condition must also be fulfilled. Sorting occurs when individuals manipulate the running variable around the cutoff to receive treatment. In this context, it would imply that drivers adjust their lap times in qualifying to secure a place in Q2 or Q3. Further elaboration

on this assumption and the reasons it is not a significant concern in this context is provided later.

3.4 Graphical Analysis

To assess whether the assumptions hold, following the approach outlined by Imbens and Lemieux (2008), a graphical analysis is required. This involves plotting the outcomes and covariates against the forcing variable to evaluate the validity of the assumptions in this specific context. Additionally, the analysis includes studying the density of the running variable.

3.4.1 Outcomes by Running Variable

The following plots analyze the average values of different outcomes at various levels of the running variable. A positive gap indicates that a driver has advanced to Q2 or Q3, while a negative gap signifies that they have been knocked out. Each plot examines whether there is a noticeable change in the outcome at the cutoff point, indicating a potential causal effect.

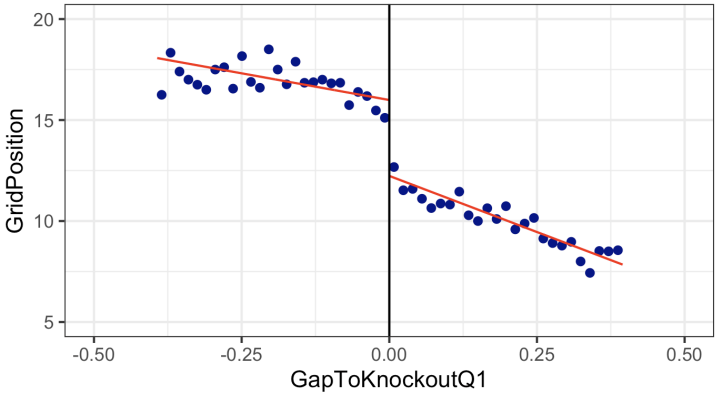

Figure 3.2. Grid Position vs. Gap to Knockout in Q1

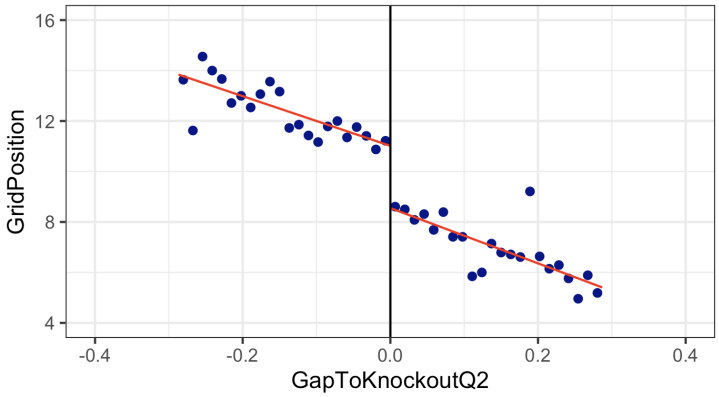

Figure 3.3. Grid Position vs. Gap to Knockout in Q2

Figures 3.2 and 3.3 display the relationship between grid position and gap to knockout in Q1 and Q2, respectively. In both cases, a clear discontinuity at the cutoff suggests a plausible causal effect of the treatment.

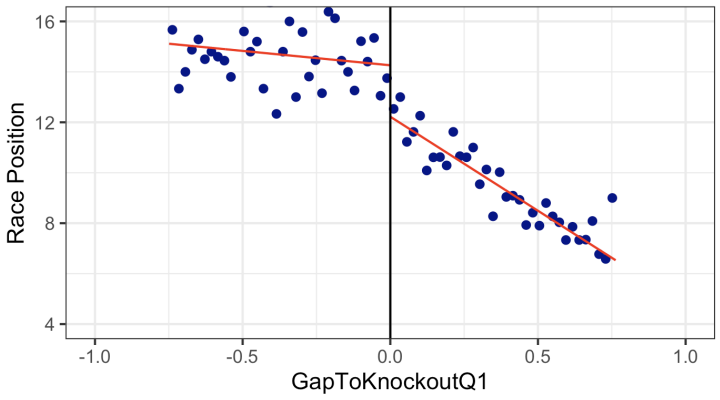

Figure 3.4. Race Position vs. Gap to Knockout in Q1

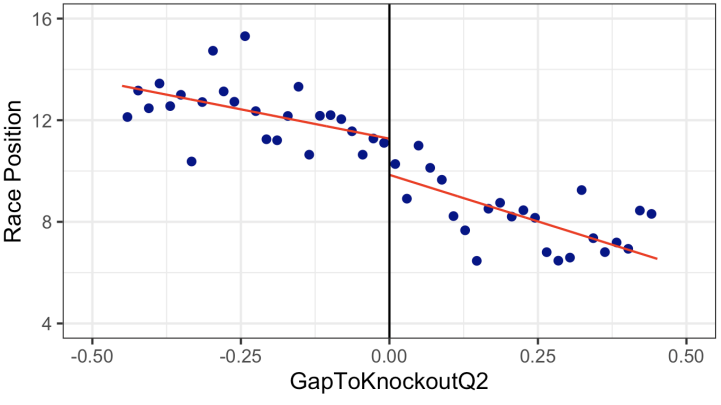

Figure 3.5. Race Position vs. Gap to Knockout in Q2

Figures 3.4 and 3.5 illustrate the relationship between race position and gap to knockout in Q1 and Q2. The observed discontinuity at the cutoff again suggests a potential effect of the treatment.

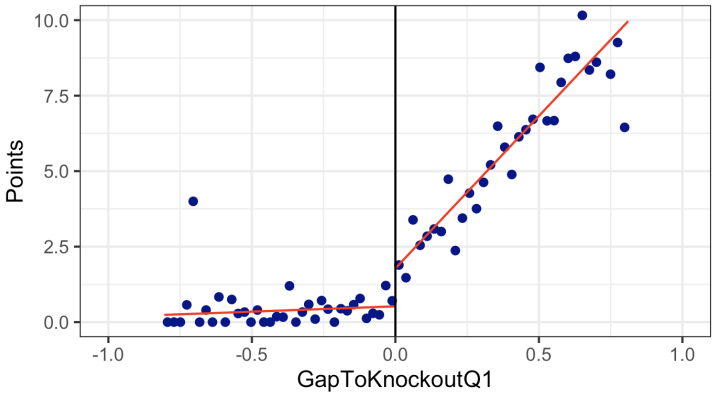

Figure 3.6. Points vs. Gap to Knockout in Q1

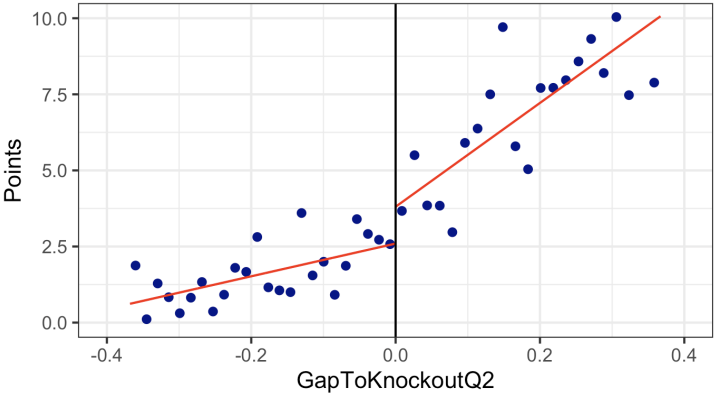

Figure 3.7. Points vs. Gap to Knockout in Q2

Figures 3.6 and 3.7 show the relationship between points and gap to knockout. The discontinuity is evident in Q1 but not in Q2, suggesting a potential treatment effect in Q1 only.

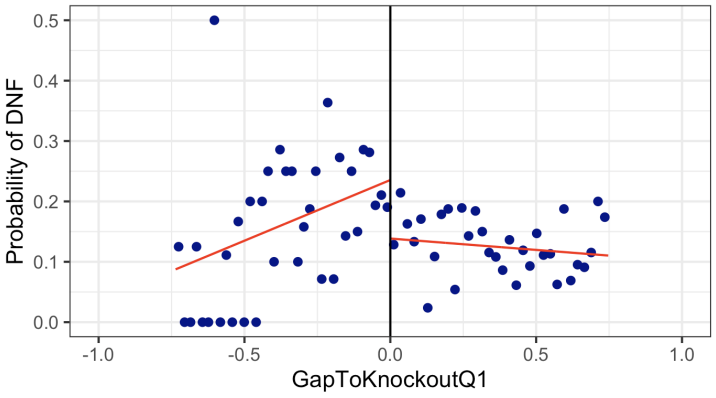

Figure 3.8. DNF Probability vs. Gap to Knockout in Q1

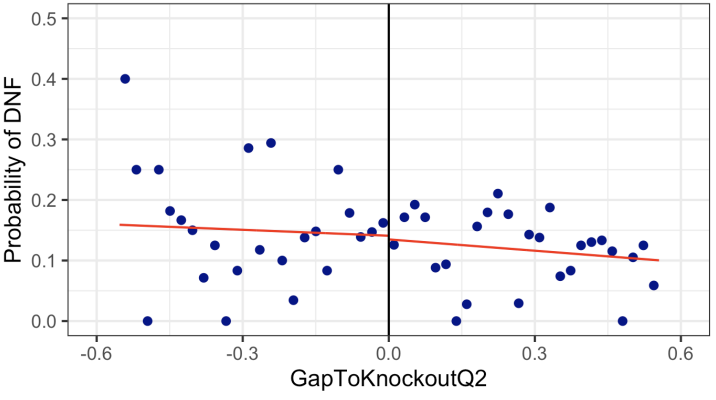

Figure 3.9. DNF Probability vs. Gap to Knockout in Q2

Figures 3.8 and 3.9 show the relationship between DNF probability and gap to knockout. The plot for Q1 shows a visible discontinuity, while Q2 does not, indicating a potential treatment effect in Q1.

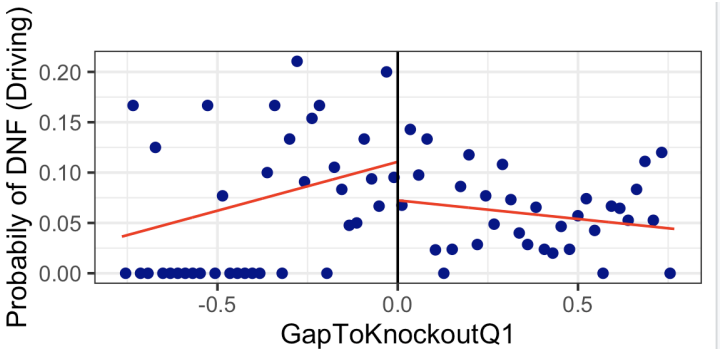

Figure 3.10. Driving DNF Probability vs. Gap to Knockout in Q1

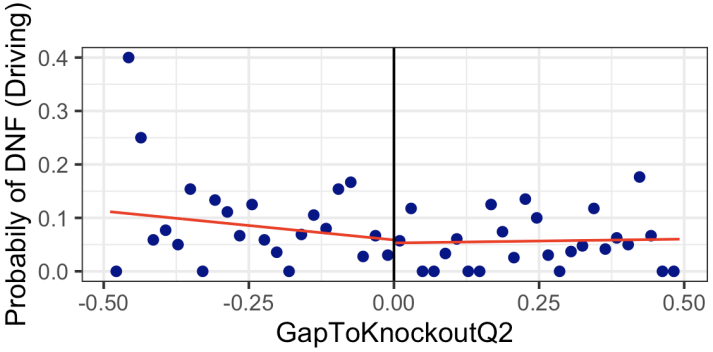

Figure 3.11. Driving DNF Probability vs. Gap to Knockout in Q2

Figures 3.10 and 3.11 illustrate the relationship between the probability of a driving DNF and the gap to knockout. The Q1 plot, has a visible discontinuity at the cutoff, whereas the Q2 plot appears continuous.

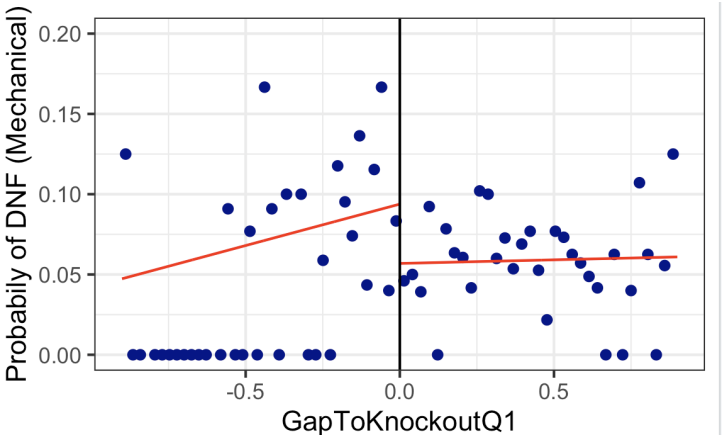

Figure 3.12. Mechanical DNF Probability vs. Gap to Knockout in Q1

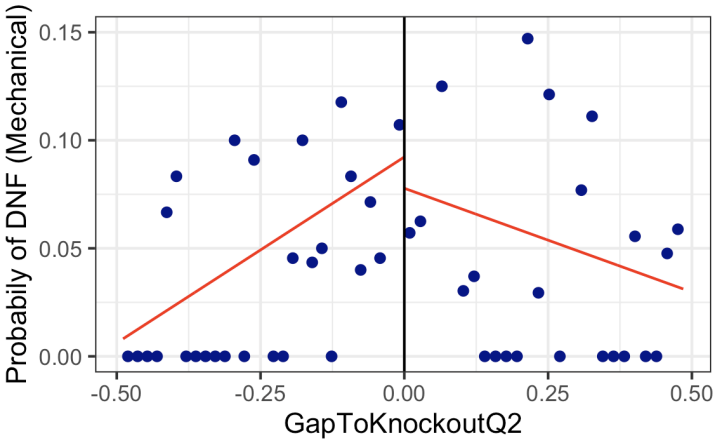

Figure 3.13. Mechanical DNF Probability vs. Gap to Knockout in Q2

Figures 3.12 and 3.13 show the relationship between the probability of a mechanical DNF and the gap to knockout. Both plots display a discontinuity, which may suggest an effect of the treatment on this

probability.

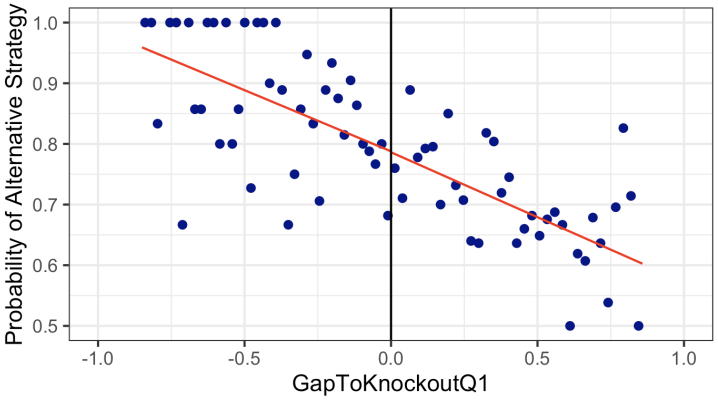

Figure 3.14. Alternative Strategy Probability vs. Gap to Knockout in Q1

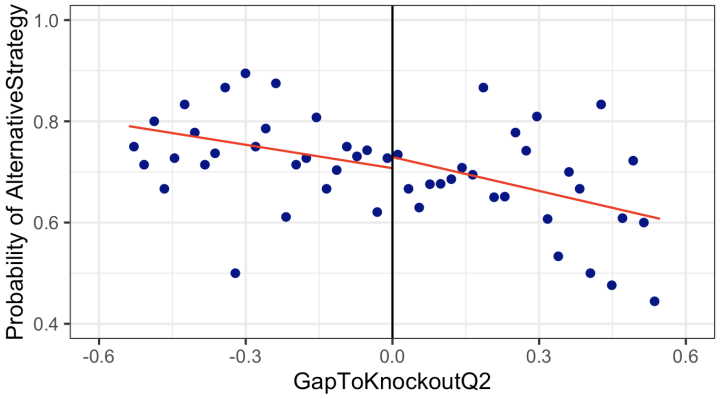

Figure 3.15. Alternative Strategy Probability vs. Gap to Knockout in Q2

Figures 3.14 and 3.15 present the probability of using an alternative strategy relative to the gap to knockout. Using an alternative strategy means employing a different approach than the one most commonly used

by the majority of drivers. No discontinuity is apparent, suggesting no significant effect on the likelihood of employing an alternative strategy.

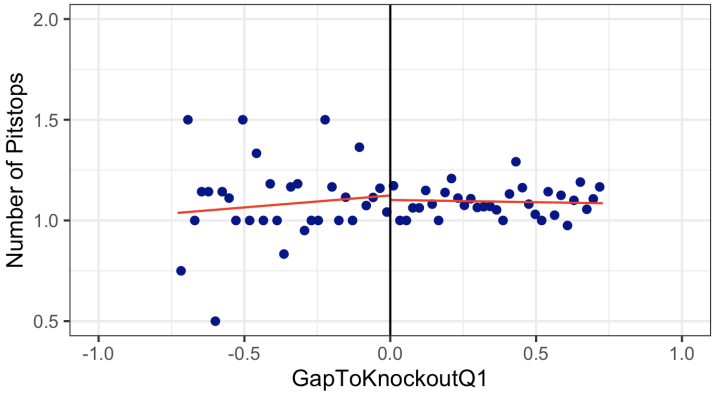

Figure 3.16. Number of Pitstops vs. Gap to Knockout in Q1

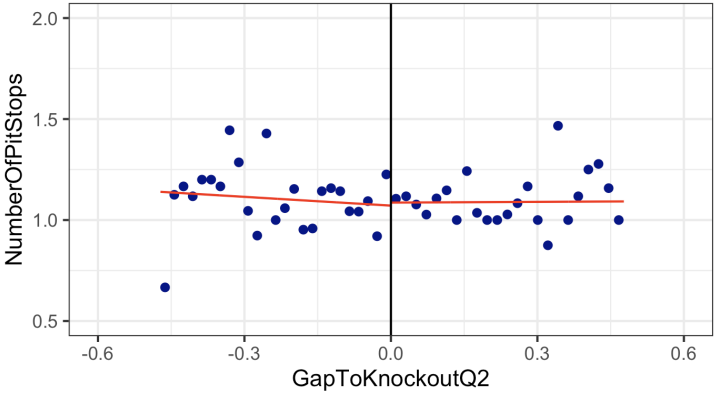

Figure 3.17. Number of Pitstops vs. Gap to Knockout in Q2

Figures 3.16 and 3.17 show the number of pitstops relative to the gap to knockout. As with the alternative strategy, no apparent discontinuity is observed, implying no significant effect on pitstops.

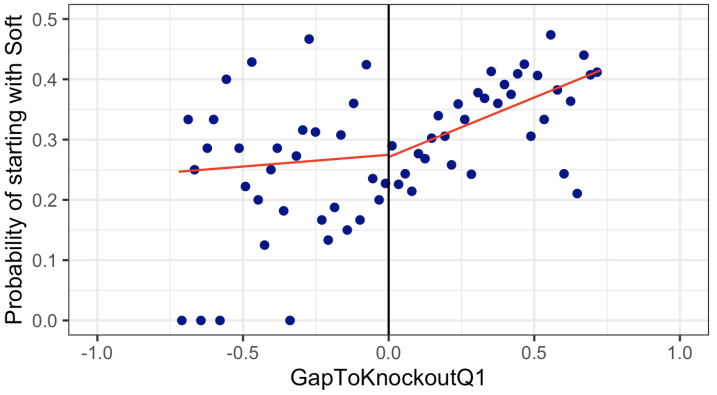

Figure 3.18. Probability of Starting with Soft Tyres vs. Gap to Knockout in Q1

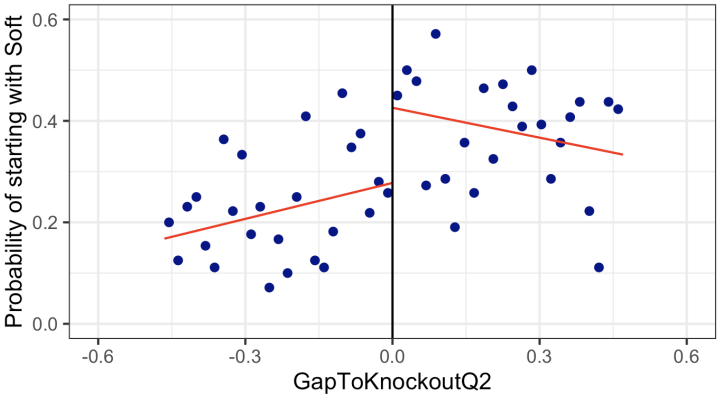

Figure 3.19. Probability of Starting with Soft Tyres vs. Gap to Knockout in Q2

Figures 3.18 and 3.19 display the probability of starting with soft tyres relative to the gap to knockout. No visual jump is observed in Q1, whereas a discontinuity is evident in Q2.

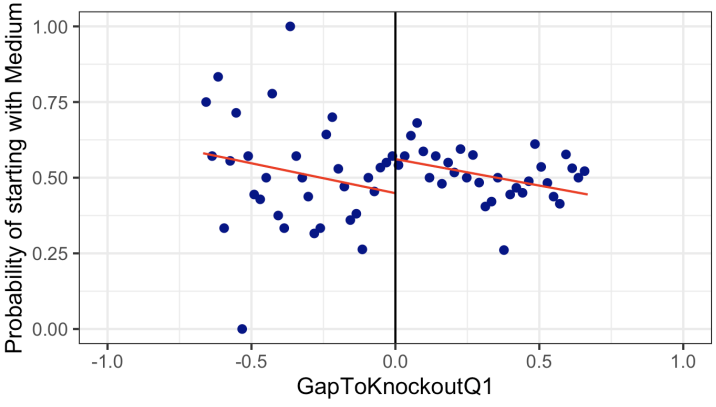

Figure 3.20. Probability of Starting with Medium Tyres vs. Gap to Knockout in Q1

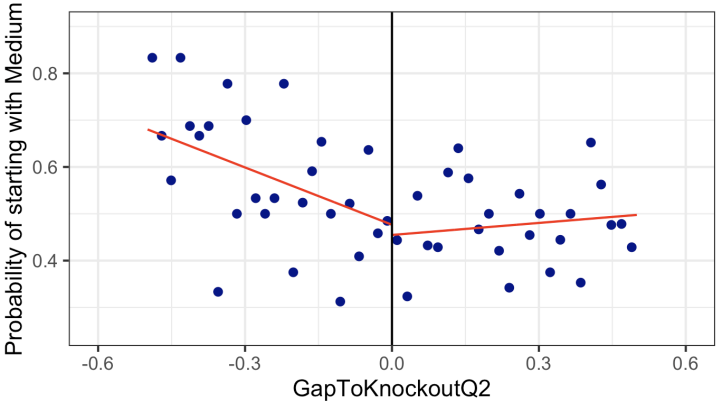

Figure 3.21. Probability of Starting with Medium Tyres vs. Gap to Knockout in Q2

Figures 3.20 and 3.21 show the probability of starting with medium tyres relative to the gap to knockout. A discontinuity is present in Q1, but it appears less pronounced in Q2.

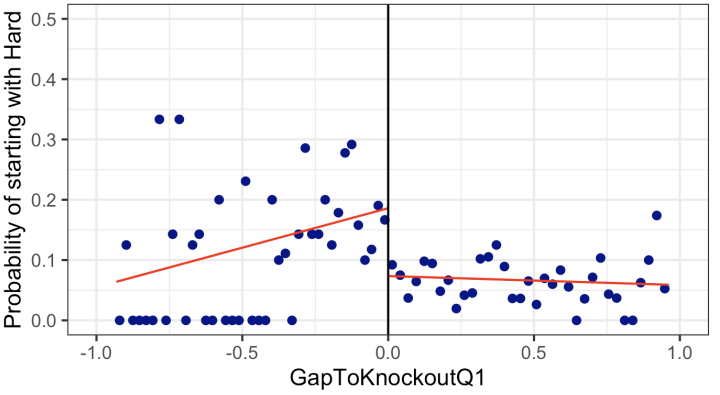

Figure 3.22. Probability of Starting with Hard Tyres vs. Gap to Knockout in Q1

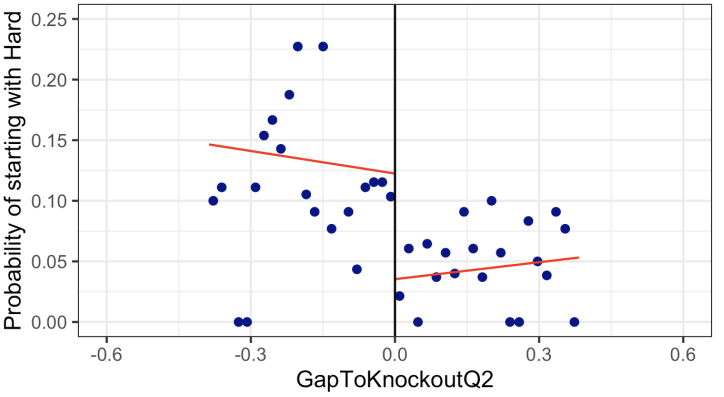

Figure 3.23. Probability of Starting with Hard Tyres vs. Gap to Knockout in Q2

Figures 3.22 and 3.23 display the probability of starting with hard tyres in relation to the gap to knockout. In both plots, a clear discontinuity suggests a potential effect of the treatment.

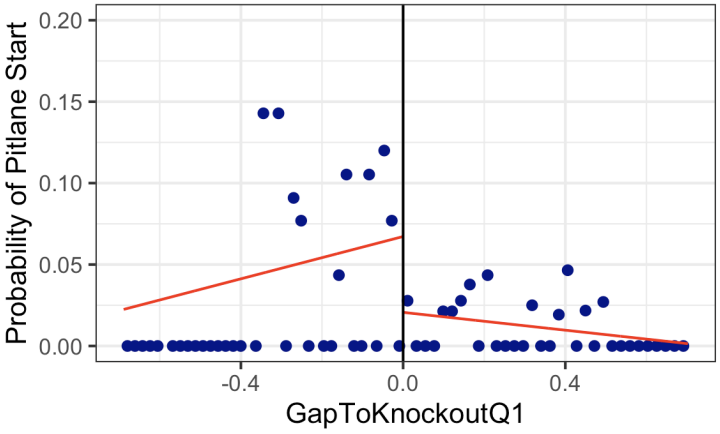

Figure 3.24. Probability of Starting from the Pitlane vs. Gap to Knockout in Q1

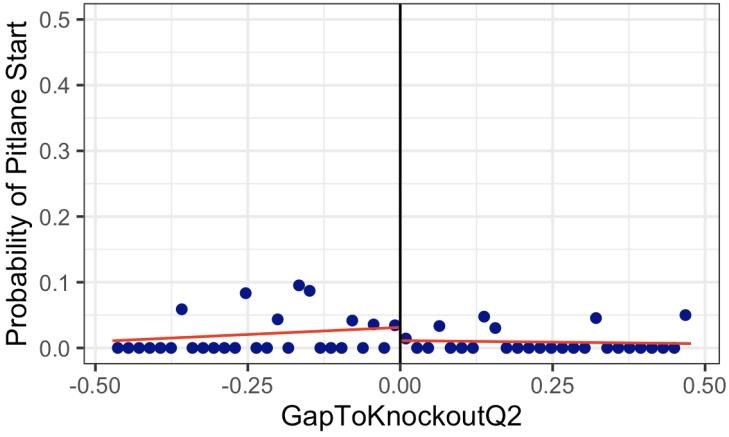

Figure 3.25. Probability of Starting from the Pitlane vs. Gap to Knockout in Q2

Figures 3.24 and 3.25 illustrate the relationship between the probability of starting from the pitlane and the gap to knockout. The Q1 plot reveals a discontinuity at the cutoff, whereas the Q2 plot

appears continuous.

3.4.2 Covariates by Running Variable

As previously mentioned, we expect no discontinuity at the cutoff for the covariates. The following plots provide visual evidence for this assertion, with additional statistical tests discussed later.

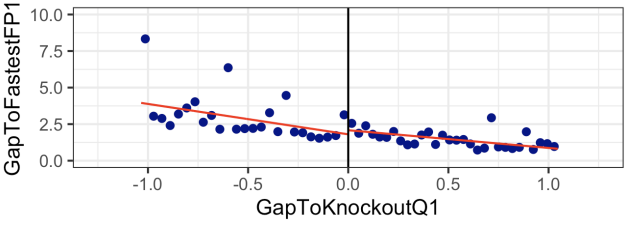

Figure 3.26. Gap to fastest in FP1 vs. Gap to Knockout in Q1

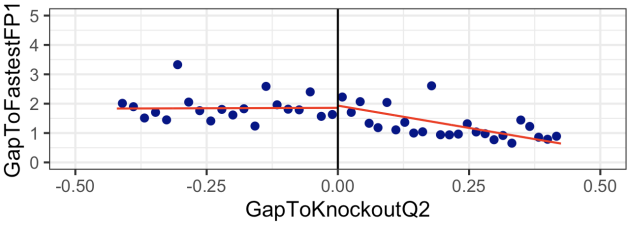

Figure 3.27. Gap to fastest in FP1 vs. Gap to Knockout in Q2

Figure 3.26 and Figure 3.27 illustrate the gap to the fastest time in Free Practice 1 against the Gap to Knockout in Q1 and Q2 sessions, showing no evident discontinuity at the cutoff.

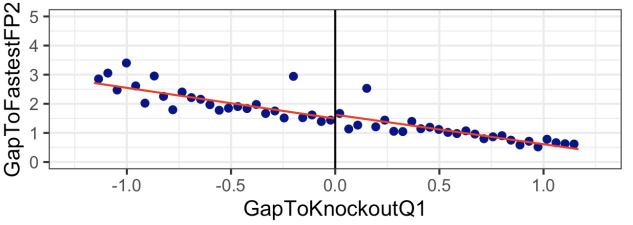

Figure 3.28. Gap to fastest in FP2 vs. Gap to Knockout in Q1

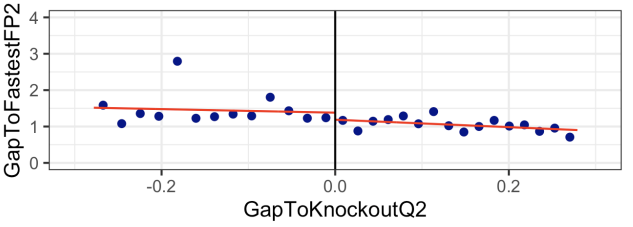

Figure 3.29. Gap to fastest in FP2 vs. Gap to Knockout in Q2

Figure 3.28 and Figure 3.29 compare the gap to the fastest time in Free Practice 2 with the Gap to Knockout in Q1 and Q2 sessions, again indicating no discontinuity at the cutoff.

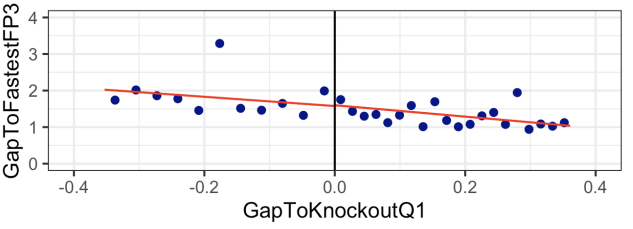

Figure 3.30. Gap to fastest in FP3 vs. Gap to Knockout in Q1

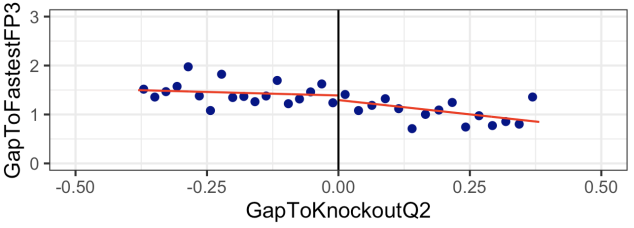

Figure 3.31. Gap to fastest in FP3 vs. Gap to Knockout in Q2

Figure 3.30 and Figure 3.31 show the gap to the fastest time in Free Practice 3 relative to the Gap to Knockout in Q1 and Q2, with no discontinuity observed at the cutoff.

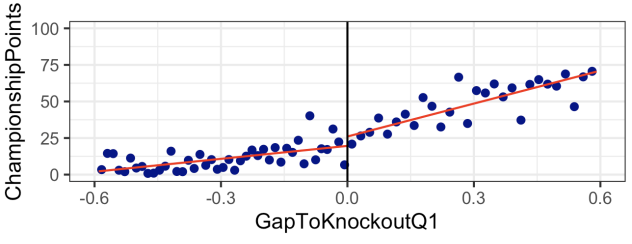

Figure 3.32. Championship points vs. Gap to Knockout in Q1

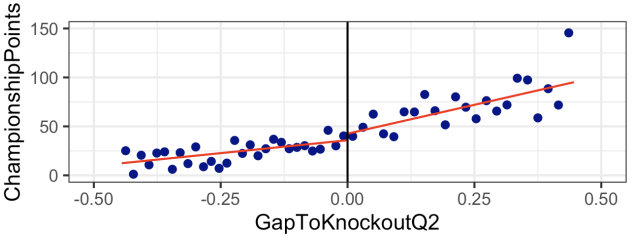

Figure 3.33. Championship points vs. Gap to Knockout in Q2

Figure 3.32 and Figure 3.33 compare Championship Points to the Gap to Knockout in Q1 and Q2, revealing no discontinuity at the cutoff.

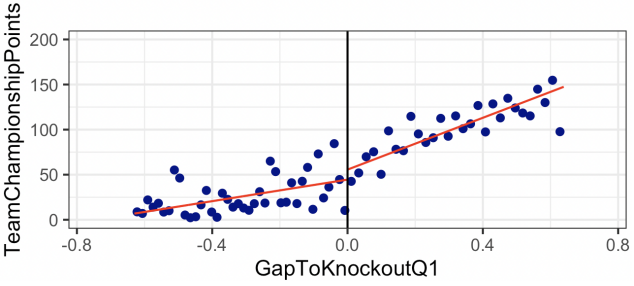

Figure 3.34. Team Championship points vs. Gap to Knockout in Q1

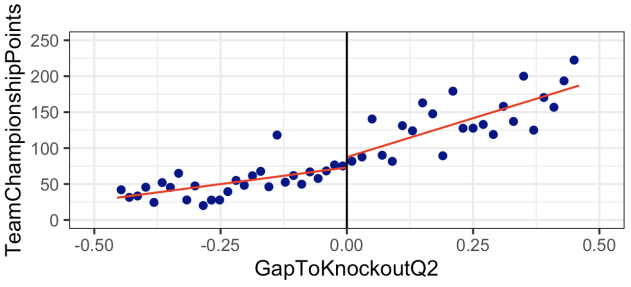

Figure 3.35. Team Championship points vs. Gap to Knockout in Q2

Figure 3.34 and Figure 3.35 show Team Championship Points in relation to the Gap to Knockout in Q1 and Q2, again with no apparent discontinuity at the cutoff.

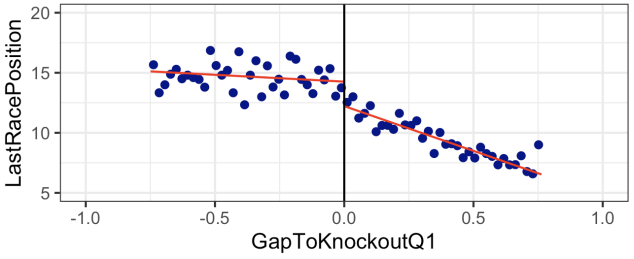

Figure 3.36. Last Race Position vs. Gap to Knockout in Q1

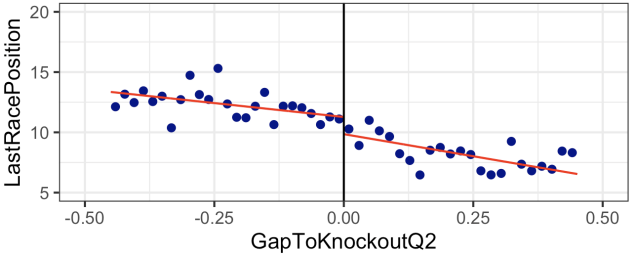

Figure 3.37. Last Race Position vs. Gap to Knockout in Q2

Figure 3.36 and Figure 3.37 compare the driver's Last Race Position to the Gap to Knockout in Q1 and Q2. The first graphic indicates a visible discontinuity, which becomes less evident in the second graphic. This issue will be addressed later with further statistical tests.

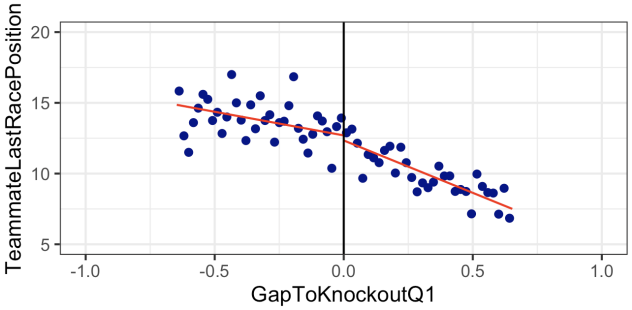

Figure 3.38. Teammate's Last Race Position vs. Gap to Knockout in Q1

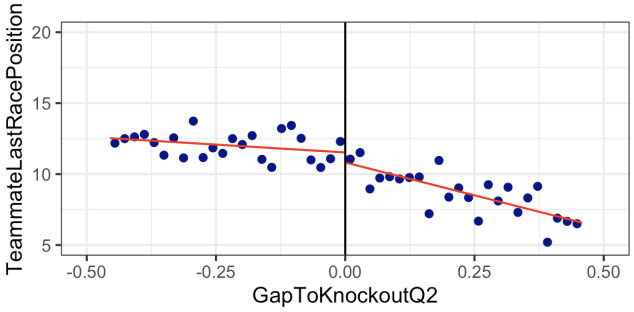

Figure 3.39. Teammate's Last Race Position vs. Gap to Knockout in Q2

Figure 3.38 and Figure 3.39 present the Teammate's Last Race Position in relation to the Gap to Knockout in Q1 and Q2, showing no apparent discontinuity at the cutoff.

3.4.3 Density of the Running Variable

Analyzing the density of the running variable helps to identify whether there has been sorting around the cutoff point. McCrary (2008) developed a test specifically for detecting manipulation in the running

variable.

Gap to Knockout	p-value
Q1	1.944582×10^{-90}
Q2	4.31126×10^{-41}

Table 3.1. McCrary Density Test p-values for Gap to Knockout in Q1 and Q2

Table 3.1 shows the p-values for the tests, which are highly significant at even the 1% level. This provides strong evidence of manipulation or sorting around the cutoff.

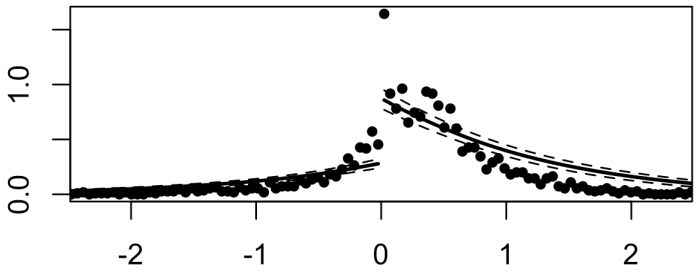

Figure 3.40. McCrary Density Test for GapToKnockoutQ1

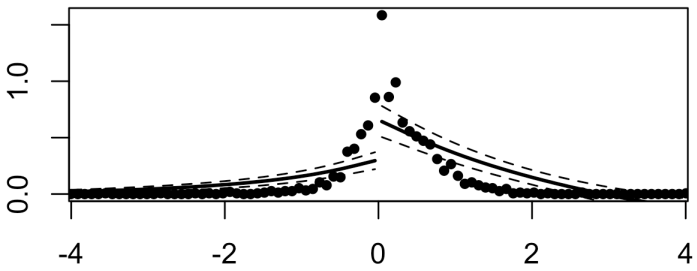

Figure 3.41. McCrary Density Test for GapToKnockoutQ2

Figures 3.40 and 3.41 display the McCrary density test results for both running variables. In each figure, there is a visible discontinuity around the cutoff, which suggests evidence of sorting or manipulation of the running variable.

However, this outcome of the test is less concerning within the context of F1 qualifying. During qualifying, teams and drivers are fully aware of the cutoff times required to advance to Q2 or Q3 and often adjust their strategies accordingly. Teams confident in their times may opt to save a set of tyres for later, while those close to the cutoff may choose to make additional fast laps on new tyres to secure advancement.

Thus, while the McCrary test indicates sorting around the cutoff, this behavior aligns with strategic decisions inherent to F1 qualifying. Consequently, the assumption of no sorting is less critical in this context.

Chapter 4

Data

4.1 Data Overview

The data used in this analysis is sourced from Formula 1's official API utilizing the Python library FastF1. This dataset encompasses information from every Grand Prix weekend from 2019 to 2024. Data from 2024 is included up to the Italian Grand Prix (Monza) (September 1st). Each record details a driver's performance throughout a race weekend, resulting in a total of 2,319 observations. The dataset comprises covariates, outcome variables, treatment variables, and running variables.

4.2 Running Variables

Running variables play a crucial role in determining the assignment of treatment. In this analysis, the running variables represent the gaps to knockout in seconds for advancing to the next session of qualifying.

The Gap to Knockout is calculated by determining the difference between a driver's lap time and the lap time of the 15th-place driver in

Table 4.1. Running Variables

Variable Name	Description	Type
GapToKnockoutQ1	Gap to the knockout time in Q1	Numeric
GapToKnockoutQ2	Gap to the knockout time in Q2	Numeric

Q1 or the 10th-place driver in Q2, respectively. A positive gap indicates that a driver has advanced to Q2 or Q3, while a negative gap signifies that they have been knocked out.

4.3 Treatment Variables

Treatment variables are indicators that determine whether a driver has progressed to the next session of qualifying.

Table 4.2. Treatment Variables

Variable Name	Description	Type
MadeItToQ2	Indicator of qualifying for Q2	Binary
MadeItToQ3	Indicator of qualifying for Q3	Binary

These are dummy variables that are set to 1 if the gap to knockout in their respective session (Q1 for MadeItToQ2 and Q2 for MadeItToQ3) is positive; otherwise, they are set to 0.

4.4 Covariates

Covariates support the continuity identifying assumption, often referred to as balance variables. They help ensure comparability between drivers around the cutoff.

Table 4.3. Covariate Variables

Variable Name	Description	Type
GapToFastestFP1	Gap to the fastest time in FP1	Numeric
GapToFastestFP2	Gap to the fastest time in FP2	Numeric
GapToFastestFP3	Gap to the fastest time in FP3	Numeric
ChampionshipPoints	Driver's accumulated points in the season	Numeric
TeamChampionshipPoints	Team's accumulated points in the season	Numeric
LastRacePosition	Driver's position in the last race	Numeric
TeammateLastRacePosition	Teammate's position in the last race	Numeric

The chosen covariates are variables that should remain stable at the cutoff, ensuring comparability among drivers. The gaps to the fastest times in Free Practices account for variations due to factors such as time of day, track evolution, and traffic, thereby reflecting the relative improvement or lack thereof compared to the fastest driver on track.

4.5 Outcomes of Interest

The effect of the treatment will be studied on these outcome variables.

Table 4.4. Outcome Variables

Variable Name	Description	Type
GridPosition	Starting grid position in the race	Numeric
RacePosition	Final position in the race	Numeric
Points	Points scored by a driver in the race	Numeric
DNF	Indicator of Did Not Finish	Binary
DNFDriving	Indicator of Did Not Finish due to a driving incident	Binary
DNFMechanical	Indicator of Did Not Finish due to a mechanical issue	Binary
StartingTyre	Tyre compound used at the start of the race	Character
NumberOfPitStops	Total number of pit stops made during the race	Numeric
AlternativeStrategy	Indicator of Alternative Race Strategy	Binary
Pitlane	Indicator of a Pitlane Start	Binary

These outcome variables will be analyzed to understand the impact of the treatment on race performance and strategy.

Chapter 5

Results

5.1 Covariate Balance Analysis

As discussed in Chapter 3, it is essential that driver characteristics near the cutoff are comparable. This implies that qualifying for Q2 or Q3 should not systematically influence the covariates if the identification assumptions hold.

To examine this, each covariate is regressed on the treatment variable (Made It to Q2 or Q3) with the same estimation equation to evaluate whether there is a statistically significant effect.

Table 5.1 presents the estimates and p-values for the effect of making it to Q2 on each covariate. The results show no statistically significant effect on any covariate except for Last Race Position, which is significant at the 1% level.

Covariate	Estimate	p-value
Gap to Fastest FP1	0.123	0.707
Gap to Fastest FP2	0.023	0.911
Gap to Fastest FP3	−0.234	0.253
Championship Points	1.631	0.625
Team Championship Points	1.951	0.716
Last Race Position	−1.259	0.01
Teammate's Last Race Position	0.278	0.514

Table 5.1. Effect of Making It to Q2 on Covariates

Covariate	Estimate	p-value
Gap to Fastest FP1	0.469	0.189
Gap to Fastest FP2	−0.186	0.372
Gap to Fastest FP3	0.009	0.938
Championship Points	0.839	0.842
Team Championship Points	2.447	0.726
Last Race Position	−0.880	0.092
Teammate's Last Race Position	0.182	0.738

Table 5.2. Effect of Making It to Q3 on Covariates

Table 5.2 shows the results for the effect of making it to Q3 on the covariates. Here, none of the covariates show a statistically significant effect, suggesting a balanced distribution on either side of the Q3 cutoff.

These findings are consistent with the RDD plots in Chapter 3, which also indicated continuity in the covariates around the cutoff. Thus, the covariate balance assumption appears to hold in this context, supporting the validity of the analysis.

5.2 Main Results: Bandwidth that Minimizes MSE

The results below show the estimated effects on the outcomes of interest within the bandwidth that minimizes the mean squared error.

	Made It to Q2	Made It to Q3
Estimate	-3.625^{***}	-1.860^{***}
Std. Error	(0.407)	(0.350)
Observations	1,129	857
Mean Grid Position: 10.53		

Table 5.3. Treatment Effect on Grid Position

Grid position determines where a driver starts in the race, with 1st being the most favorable and 20th the least. As shown in Table 5.3, both treatment estimates are negative, indicating that drivers who make it to Q2 or Q3 secure starting positions closer to the front. Specifically, advancing to Q2 leads to an estimated improvement of 3.6 positions, and advancing to Q3 results in an improvement of approximately 1.9 positions. These significant estimates affirm that qualifying progress positively impacts starting grid positions, highlighting the strategic importance teams place on advancing through qualifying rounds.

	Made It to Q2	Made It to Q3
Estimate	-2.022^{***}	-0.942^{*}
Std. Error	(0.431)	(0.548)
Observations	1,728	1,117
Mean Race Position: 10.49		

Table 5.4. Treatment Effect on Race Position

Race position represents the final order in which drivers finish, with the top ten scoring points. In Table 5.4, both estimates are again negative, indicating a move toward more favorable race finishing positions for those who qualify for Q2 or Q3. Specifically, advancing to Q2 leads to an improvement of about 2 positions in the final race order, while advancing to Q3 yields an improvement of nearly 1 position. This positive effect demonstrates that qualifying performance influences race outcomes by securing better finishing positions.

	Made It to Q2	Made It to Q3
Estimate	1.043^{**}	0.921
Std. Error	(0.456)	(0.621)
Observations	1,777	999
Mean Points: 5.06		

Table 5.5. Treatment Effect on Points

Points represent the number of points a driver scores by the end of the race weekend. As seen in Table 5.5, advancing to Q2 corresponds to an increase of about 1 point, showing a meaningful improvement in overall scoring potential. However, the effect of making it to Q3 is not statistically significant. Since Q3 drivers already start in the points-paying positions, this lack of significance may indicate

diminishing marginal gains from advancing further in qualifying for scoring purposes.

	Made It to Q2	Made It to Q3
DNF		
Estimate	-0.096^{***}	0.014
Std. Error	(0.035)	(0.037)
Observations	1,751	1,258
Mean DNF: 0.14		
Driving DNF		
Estimate	-0.028	0.019
Std. Error	(0.024)	(0.027)
Observations	1,773	1,193
Mean Driving DNF: 0.070		
Mechanical DNF		
Estimate	-0.050^{**}	-0.020
Std. Error	(0.023)	(0.028)
Observations	1,880	1,192
Mean Mechanical DNF: 0.062		

Table 5.6. Treatment Effect on DNF Types

The probability of DNF (Did Not Finish) refers to the likelihood of a driver failing to complete the race. Table 5.6 shows a significant reduction in the overall DNF probability of 0.096 for drivers advancing to Q2, indicating increased reliability or race completion for these drivers. However, the effect of advancing to Q3 is not statistically significant, suggesting that reaching Q3 alone does not further impact the likelihood of race completion.

The DNF outcome is divided into two sub-outcomes: Driving DNF

and Mechanical DNF. Driving DNF refers to situations where a driver cannot complete the race due to incidents attributable to driving, such as accidents, collisions, running over debris, punctures, etc. Mechanical DNF, on the other hand, refers to failures within the car, such as issues with the gearbox or hydraulics, for example.

For Driving DNF, there are no significant effects of either treatment, while advancing to Q2 reduces the probability of a Mechanical DNF by 0.050.

	Made It to Q2	Made It to Q3
AlternativeStrategy		
Estimate	−0.017	0.029
Std. Error	(0.041)	(0.050)
Observations	1,847	1,250
Mean Alternative Strategy: 0.72		
NumberOfPitStops		
Estimate	−0.017	0.008
Std. Error	(0.051)	(0.058)
Observations	1,729	1,180
Mean Number of Pitstops: 1.08		

Table 5.7. Treatment Effect on Tyre Strategy

Table 5.7 examines the impact on tyre strategy, specifically alternative strategies and the number of pit stops. Neither advancing to Q2 nor Q3 shows a significant effect on adopting alternative strategies or changing the number of pit stops. This may indicate that qualifying position does not drastically influence a team's tyre strategy decisions as a full strategy.

	Made It to Q2	Made It to Q3
TyreSOFT		
Estimate	−0.014	0.126***
Std. Error	(0.041)	(0.049)
Observations	1,725	1,174
Mean TyreSOFT: 0.30		
TyreMEDIUM		
Estimate	0.122**	−0.030
Std. Error	(0.048)	(0.054)
Observations	1,657	1,203
Mean TyreMEDIUM: 0.51		
TyreHARD		
Estimate	−0.124***	−0.052
Std. Error	(0.025)	(0.032)
Observations	1,933	1,052
Mean TyreHARD: 0.08		

Table 5.8. Treatment Effect on Starting Tyre

Table 5.8 shows the treatment effects on the likelihood of using specific tyre compounds at the start. Advancing to Q2 increases the probability of starting on medium tyres, while reducing the likelihood of starting on hard tyres, possibly due to teams aiming for a balance between durability and performance. For drivers who make it to Q3, the significant positive effect on starting with soft tyres may reflect a strategic choice to secure the best possible race start, emphasizing tyre grip over durability.

	MadeItToQ2	MadeItToQ3
Estimate	-0.056^{***}	-0.004
Std. Error	(0.015)	(0.014)
Observations	1,700	1,178
Mean Pitlane Start: 0.025		

Table 5.9. Treatment Effect on Probability of Pitlane Start

Table 5.9 shows the treatment effects on the probability of a pitlane start. Drivers are required to start from the pitlane if their car has been modified after qualifying, which is a penalty applied due to a breach of Parc Fermé conditions. Advancing to Q2 reduces the probability of starting from the pitlane by 0.056. This is logical because if teams are knocked out in Q1, they may choose to take the penalty to change the car setup and lose only a few positions, as compared to advancing to Q3, which has no significant effect on the probability of a pitlane start when drivers are further up the grid.

5.3 Robustness Analysis

This section presents a robustness analysis to assess the reliability and consistency of the results obtained from the main analysis. Varying the bandwidth and artificially shifting the cutoff are both effective methods for testing the robustness of the results.

5.3.1 Different Bandwidths

The robustness of the main results is tested by using different bandwidths, ranging from 0.05 to 1.05 seconds in increments of 0.10. It is expected that the significance of the results will remain consistent

across these different bandwidths, indicating that the findings are robust to the choice of bandwidth.

The outcomes for Grid and Race positions are largely robust to bandwidth choice in both Made it to Q2 and Made it to Q3 treatments. The only exception is the 0.05 bandwidth, where no significant treatment effect is observed.

For the DNF outcome, the results for Made it to Q2 show robustness similar to the position outcomes, with only the 0.05 bandwidth showing no significant treatment effect. For Made it to Q3, all bandwidths suggest no significant effect, aligning with the main results and indicating robustness. In contrast, for the DNF sub-outcomes, the effect on Driving DNF is robust. However, the effect of making it to Q2 on the probability of a Mechanical DNF appears less robust, as shown in Figure 3.13. Less than half of the confidence intervals for different bandwidths suggest a significant effect.

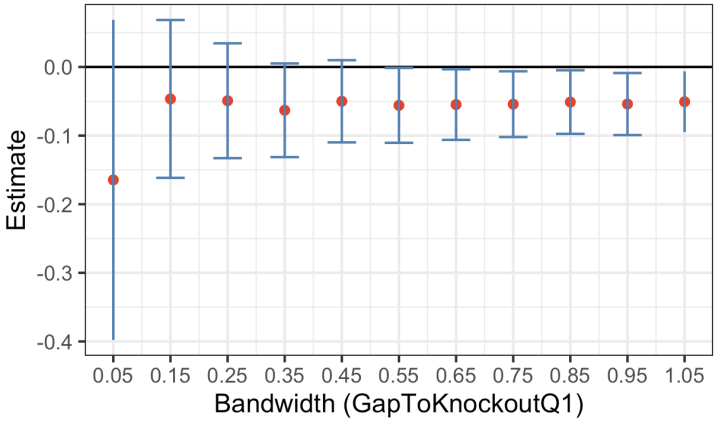

Figure 5.1. Confidence intervals for the effect of Made It To Q2 on Mechanical DNF probability

The treatment effect on Points is robust for Made it to Q2, but for Made it to Q3, Figure 5.2 shows that nearly half of the bandwidths indicate a significant effect, even though the main results suggest no significant effect. This may indicate that the treatment effect on this outcome is less robust.

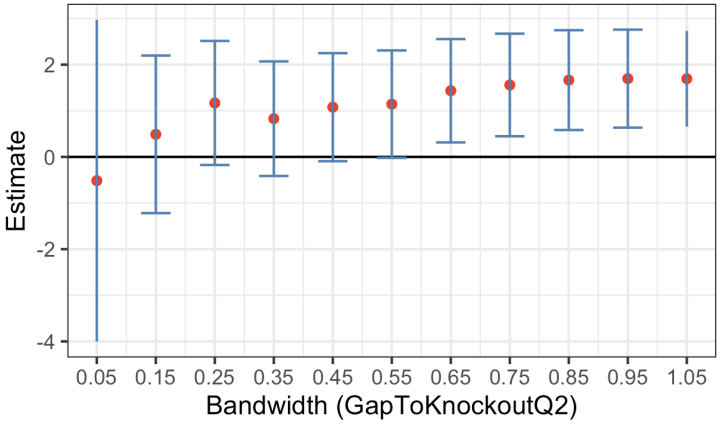

Figure 5.2. Confidence intervals for the effect of Made It To Q3 on Points

The treatment effects on Alternative Strategy and Number of Pitstops are robust, as all bandwidths suggest the same conclusion as the main results: there is no significant effect.

The treatment effects on starting tyre are also robust. However, the main results suggested that the effect of making it to Q2 on starting with Mediums is significant. As observed in Figure 5.3, less than half of the bandwidths suggest this, so this result may not be as robust. The effect on Pitlane start is also robust for both treatments.

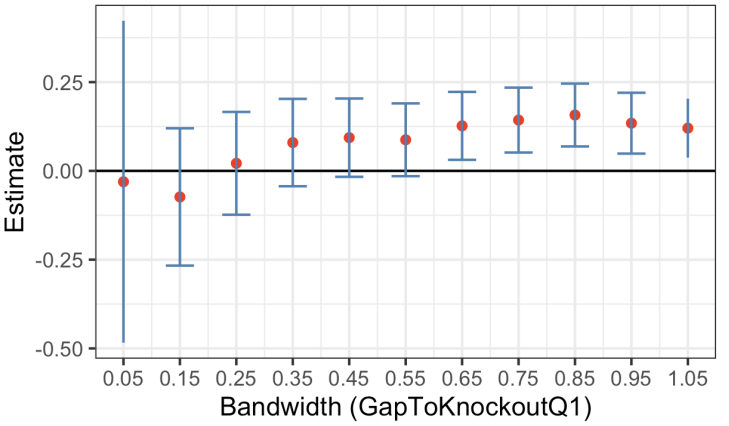

Figure 5.3. Confidence intervals for the effect of Made It To Q2 on starting with Mediums

5.3.2 Different Cutoffs

To further evaluate robustness, the cutoff is artificially shifted. If the results are robust, only the actual cutoff should produce a significant effect, if any, on the outcomes. For this test, five cutoff values are examined: -1, -0.5, 0 (the actual cutoff), 0.5, and 1.

Overall, the treatment effects of both Made it to Q2 and Made it to Q3 are robust to changes in the cutoff. Where the main results indicated a significant treatment effect, the confidence intervals for alternative cutoffs do not suggest significant effects, except at the actual cutoff. For outcomes without a significant effect in the main results, neither the actual cutoff nor the artificial cutoffs show a significant effect.

It is worth noting that there were cases of significance at alternate cutoffs: the effect of Made it to Q2 on Grid Position shows significance at a cutoff of -0.5, as seen in Figure 5.4. Similarly, the effect of Made it to Q2 on the probability of starting with Hard tyres shows significance at the same cutoff (-0.5), as displayed in Figure 5.5.

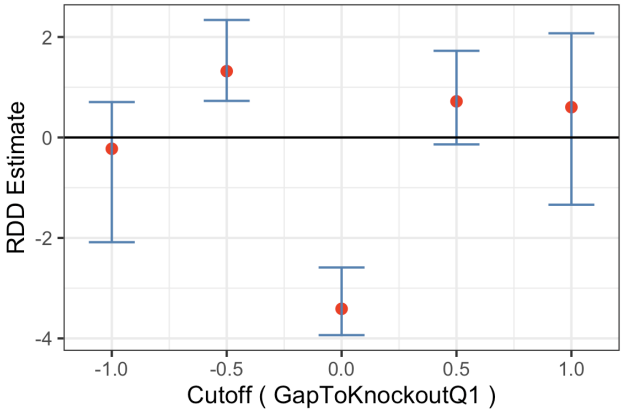

Figure 5.4. Confidence intervals for the effect of Made It To Q2 on Grid Position

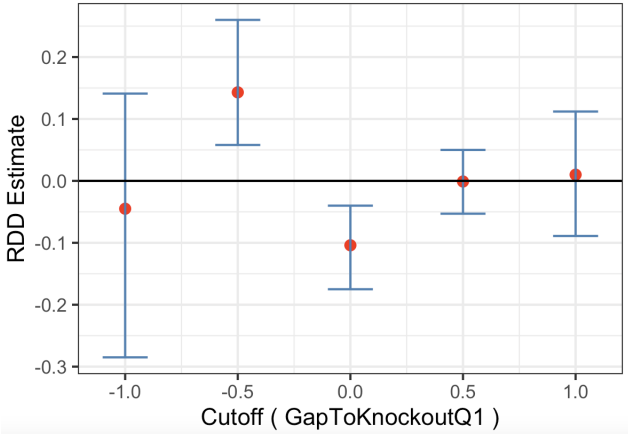

Figure 5.5. Confidence intervals for the effect of Made It To Q2 on probability of starting with Hard tyres

Conclusions

This paper addresses the question of whether the belief that qualifying can make or break a driver's Sunday holds true. The analysis demonstrates that advancing to Q2 and Q3 has a significant impact on various race outcomes.

Using a Regression Discontinuity Design (RDD), the study finds robust qualifying effects on most race outcomes. Advancing to Q2 generally has the largest impact: drivers who make it through to this stage improve their starting position by approximately 3.6 places and their final race position by around 2 places. While advancing to Q3 also has positive effects, these are smaller, improving Grid Position by 1.9 places and Final Race Position by about 1 place.

Interestingly, the analysis identifies outcomes where advancing to Q3 does not yield significant effects, such as points, the probability of a DNF, or the likelihood of starting from the pitlane. Additionally, the lack of significant effects on the adoption of alternative strategies is unexpected. Typically, teams starting further back or from the pitlane adopt different tyre strategies to recover ground lost during qualifying. This finding suggests that the impact of alternative strategies may depend on factors beyond qualifying performance, such as in-race developments or external conditions.

The implications for Formula 1 teams are clear: prioritizing

advancement to Q2 should be a strategic focus, especially for teams on the edge of elimination in Q1. Teams in the midfield or toward the back of the field may benefit more from concentrating their efforts on securing a place in Q2 rather than stretching to achieve Q3 qualification. Advancing to Q2 provides the largest performance gains and sets the foundation for stronger race results.

However, the results also highlight the role of external factors in determining race outcomes. Teams that marginally make it to Q2 or Q3 may be aided or hindered by events beyond their control, such as yellow or red flags, mechanical failures, or errors by other drivers. These factors introduce an element of unpredictability—akin to "qualifying luck"—that can significantly influence results.

This concept of "luck" extends beyond Formula 1. Similar to how exam luck influences educational and labor market outcomes as exogenous variation in competitive environments can alter trajectories. For instance, individuals awarded scholarships based on performance metrics affected by random variation may experience advantages that shape their future opportunities. Both in Formula 1 and outside it, randomness and external factors play a crucial role in determining outcomes, highlighting the interplay between preparation, performance, and chance.

In conclusion, this study shows that qualifying performance plays a key role in shaping race outcomes, with advancing to Q2 having the most significant impact. Teams that make it to Q2 can expect better race results, and for those near the edge, focusing on Q2 qualification should be their main goal, even if it means not reaching Q3.

However, the analysis also highlights the unpredictable nature of racing. External factors like yellow or red flags, technical issues, or mistakes from other drivers can affect the qualifying result. This mirrors

how random factors, such as exam luck, can influence people's success in areas beyond racing.

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